Step 1—Choose Implementation Benchmark

The first step of the macro decision process is selection of the implementation price benchmark. The more common price benchmarks include implementation shortfall (IS), decision price, price at order entry ("inline"), opening price, prior night's close, future closing price, and VWAP. Another common implementation goal is to minimize tracking error compared to some benchmark index. It is essential that the implementation goal be consistent with the manager's investment decision. For example, a value manager may desire execution at their decision price (i.e., the price used in the portfolio construction phase), a mutual fund manager may desire execution at the closing price to coincide with valuation of the fund, and an indexer may desire execution that achieves VWAP (e.g., to minimize market impact) or one that minimizes tracking error to their benchmark index.

Step 2—Select Optimal Execution Strategy

The second step of the process consists of determining the appropriate optimal execution strategy. This step is most often based on transaction cost analysis. Investors typically spend enormous resources estimating stock alphas and consider these models proprietary. Market impact estimates, however, remains the holy grail of transaction cost analysis and are usually provided by brokers due to the large quantity of data required for robust estimation. Risk estimates, on the other hand, can be supplied by investors, brokers, or a third party firm.

The selected optimal execution strategy could be defined in terms of a trade schedule (also referred to as slicing strategy, trade trajectory, waves), a percentage of volume ("POV"), as well as other types of liquidity participation or price target strategies. For example, trade as much as possible at a specified price or better.

The more advanced investor's will perform TCA optimization. This consists of running a cost-risk optimization where the cost component consist of price trend, and market impact cost. That is,

$$Min \quad (MI + Trend) + \lambda \cdot Risk$$

where λ is the investor's specified level of risk aversion. Investors who are more risk averse will set $\lambda > 1$ and investors who are less risk averse will set $\lambda < 1$. In situations where the trader does not have any expectations regarding price trend, our TCA optimization is written in terms of market impact cost and risk as follows:

$$Min \quad MI + \lambda \cdot Risk$$

Depending upon expected price trend, optimization may determine an appropriate front- and/or back-loading algorithm to take advantage of